

Field-Circuit Coupling Using Black-Box Simulation Tools

Coupling to a Reduced-Order Field Representation via Waveform Relaxation

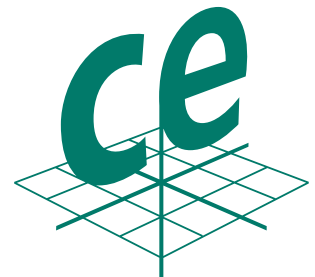
Bachelor thesis in the field of study "Computational Engineering" by Philip Simon

Date of submission: July 31, 2026

1. Review: Gutachter 1
2. Review: Gutachterin 2
Darmstadt



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field of study:
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Bei einer Thesis des Fachbereichs Architektur entspricht die eingereichte elektronische Fassung dem vorgestellten Modell und den vorgelegten Plänen.

Darmstadt, 31. Juli 2026

P. Simon

Abstract

Electric machines operate at relatively low frequencies yet are typically driven by high-frequency power-electronic circuitry, so they are most efficiently time-integrated at different rates. Their differing character – one a complex machine, modelled as a distributed device with differential equations and simulated numerically using methods such as the Finite Element Method (FEM), the other a circuit solved by its own solver – motivates a partitioned co-simulation rather than a monolithic solve. This work couples the domains with a Waveform Relaxation (WR) scheme, in which an inductance-based reduced-order model of the field serves as the interface condition and, as a linearisation term, accelerates convergence. The scheme is implemented with Xyce, Sandia's open-source SPICE simulator, leveraging its checkpoint/restart feature to re-solve each time window across WR iterations. The implementation is verified against monolithic solutions for cases with varying circuit complexity and a dummy FEM solver. A parameter study over [window length/linearisation update / ...] shows that [key result], from which [a concrete guideline, e.g. for choosing the window size] is derived.

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1 Einleitung

1.1 Motivation

Most parts of a drive system, such as sources, switches, passive elements, and control circuitry, do not need to be modelled with the full spatial detail of Maxwell's partial differential equations [12] and can be adequately described by lumped elements. These reduced models are formulated as differential-algebraic equations (DAEs), e.g. by modified nodal analysis (MNA) [6]. For the machine itself, a lumped representation may be insufficient whenever saturation, eddy-current effects or the spatial distribution of the field govern its behaviour. A spatially distributed model is used instead, commonly obtained from the magnetoquasistatic (MQS) approximation of Maxwell's equations and discretised in space, e.g. by the finite element method (FEM) [7]. Coupling such a field model to the circuit yields a system of partial differential-algebraic equations (PDAEs), which reduces to a system of DAEs once the field model is discretised in space [3]. Solving the system *monolithically* would require a single time integrator to resolve the dynamics of the most active domain, so the latent domain is advanced with a step size far smaller than its dynamics require. Partitioning removes this restriction, as each subsystem can be integrated with a step size adapted to its own time constants. This is particularly beneficial when the time scales of the domains are separated by orders of magnitude, where it has been shown to reduce the computational effort considerably [11]. Electric machines are increasingly fed by fast-switching power electronics, enabled by wide-bandgap semiconductor devices and driven by ambitious power-density and voltage targets for the coming decade, see Tab. 1.1. In such systems, the machine is the latent subsystem, as its electromagnetic and mechanical time constants exceed the converter switching period by orders of magnitude, which motivates multirate and co-simulation approaches [10, 8]. Faster switching also increases the voltage slew rate dv/dt at the machine terminals. Whereas insulation systems of standard machines are designed for slew rates of a few $\text{kV}\mu\text{s}^{-1}$, silicon-carbide devices reach values exceeding $100\text{ kV}\mu\text{s}^{-1}$ [9]. Consequently, the behaviour of the machine at its terminals is no longer adequately represented by a lumped element, motivating a partitioned field/circuit approach in which the interface conditions capture the machine's terminal response.

Practically speaking, there are well-established solvers for each domain. However, commercial software is often closed-source and access to the internal variables during simulation is limited, making coupling the field/circuit domains difficult, cf. [5]. Hence, a so-called *black-box* view of the problem, which is agnostic to the solvers, is the versatile implementation. More precisely, one might call it a *grey-box* view, since we require some knowledge of the state of the field domain in the interface condition, as we will discuss later.

1.2 State of the Art

Solving the system of DAEs is a thoroughly studied problem in the context of simulating field/circuit domains. Approaches differ in how tightly the subsystems are coupled, ranging from monolithic solves, where all

Table 1.1: Power electronics and electric motor targets (US DoE).

	2025	2030	2035
Voltage (V)	600	800	800
Power Density Electronics (kW/L)	150	200	225
Power Density Machines (kW/L)	75	100	112,5

Daten aus [13].

equations are assembled into a single system, to loosely coupled *dynamic iteration* (also called *Waveform Relaxation* (WR)) where the subsystems are solved separately and exchange waveforms over time windows, see Section ???. The monolithic approach requires a shared solver for both domains, which conflicts with the black-box setting and is prohibitively expensive in computation for the targeted application here, see Section 1.1. In the case of dynamic iteration, it has been identified that the exchange of algebraic variables is a mathematical cause for divergence [10]. Thus, enhanced modelling of the interface via a *surrogate* or *reduced order model* (ROM) has been shown to improve convergence significantly. Without it, the iteration may be divergent [2]. While such an interface condition is straightforward to formulate analytically, its time discretisation involves the time integrators of both subsystems. When black-box tools are coupled, each subsystem discretises the shared condition with its own integrator, so the two discrete operators may differ. Different operators result in an inconsistent interface condition and may lead to a false fixed point or a divergent iteration, as we will see in Chapter 4. This motivates the need for a consistent implementation of the interface without access to the solver’s time integrator. Xyce™ also provides a native coupling interface (General External Interface). Why it is not used for the dynamic iteration here is discussed in Section 3.4.

1.3 Objective and Contributions

The contributions of this thesis are:

1. a dynamic iteration co-simulation of the field/circuit built entirely from black-box, netlist-driven simulators (demonstrated with Xyce™);
2. a consistent formulation and implementation of the interface condition that does not require access to either solver’s time integrator;
3. a study of the resulting scheme’s convergence behaviour, transmission conditions and parameter-dependence, e.g. window-length, isolated from any particular field discretisation.

Xyce™ is the open-source analogue circuit simulator used in this thesis. It can be replaced by other netlist-driven simulators with little change to the implementation, provided they offer:

- Restart/Checkpointing capability
- Behavioural source-elements

1.4 Limitations

The coupling scheme developed in this work makes no assumption about how the field model is formulated or discretised internally, at the cost of not demonstrating the method on an industrial machine model. The latter is left to future work. Considering anything other than two-port distributed field devices would require substantial changes to the codebase. The ROM values are constant across the simulation, such that changes of state in the distributed device, e.g. by saturation, are not accurately modelled by the ROM. The implementation only covers transient simulation.

1.5 Outline of the Thesis

In Chapter 2, the fundamental theory that is required to explain the implementation and discuss any results is laid out. Chapter 3 documents the implementation of the co-simulation with the Xyce[™] parallel electronic simulator. In Chapter 4, the implementation is verified against a suite of benchmark problems. Chapter 5 contains parameter studies and a general approach to choosing them. Chapter 6 concludes the thesis and outlines possibilities for future work.

2 Fundamentals

One of the most significant ideas that improved simulation speed by up to two orders of magnitude is decomposing large systems into coupled subsystems that can be integrated independently [5]. To implement the coupling of the field and circuit domains via dynamic iteration, the underlying concepts need to be explained first.

2.1 Dynamic Iteration

The subsystems are integrated independently for a period of time $T_n \rightarrow T_{n+1}$, where $T_n = [t_n, t_{n+1}]$, using the outputs of neighbouring subsystems as inputs. This is the essence of dynamic iteration [5]. It is a fixed-point iteration technique, i.e., the iteration

$$x^{k+1} = \varphi(x^k), \quad k = 0, 1, 2, \dots$$

converges to a fixed point, where $x^* = \varphi(x^*)$.

In the case of a field-circuit coupling, the vector of unknowns x contains the voltage (node potentials) and (branch) current across the interface ports. The fixed point is reached if the voltages and currents of both subsystems converge, i.e.,

$$\begin{array}{cc} V_C^{k+1} = V_C^k & V_F^{k+1} = V_F^k \\ \underbrace{I_C^{k+1} = I_C^k}_{\text{circuit}} & \underbrace{I_F^{k+1} = I_F^k}_{\text{field}} \end{array} \quad (2.1)$$

where V_C and I_C are the port voltage and current of the circuit-side, and V_F and I_F are the port voltage and current of the field-side. dd

2.2 Coupling Scheme

Transmission between the subsystems is handled at the ports, i.e., the inputs and outputs of a subsystem. Two independent factors determine the coupling of the field and circuit: the coupling direction and the transmission condition itself.

Coupling Direction The direction of coupling stipulates which variable from one subsystem (output) is prescribed on another (input) during iteration. For our field/circuit co-simulation, it answers the question: *“Does the field ingest a current or a voltage as its (plain) input?”*. This consideration directly impacts the index of the resulting DAE. It has been shown that voltage-driven field formulations (A^* and $T\text{-}\Omega$) result in a lower index system and thereby offer better numerical stability than the current-driven alternative [1, 3]. Hence, a voltage-driven coupling

$$V_F \leftarrow V_C$$

is likely the preferred choice for this work’s implementation, see Section 3, though under some scopes the current-driven approach may be considered, cf. e.g. [4].

Transmission Conditions Also called the interface conditions, determine *how* the variables are used as inputs and outputs of the subsystems. In classical WR, the plain (zero-order) transmission conditions of the field/circuit problem are

$$V_F \leftarrow V_C \quad \text{and} \quad I_C \leftarrow I_F, \quad (2.2)$$

in case of a voltage-driven coupling. Note that the exchanged variables are waveforms, i.e., $V_C = V_C(t)$, hence the name “Waveform Relaxation”. Previous work [11, 10] has shown the benefit of extracting a ROM of the field domain. The better the ROM, the larger the time windows can be chosen. Since MQS formulations of the field are inherently inductive, so is the ROM [3]. Later work in [4] showed how the ROM is really just the first-order approximation to the optimal transmission condition.

Following the suggestion of [4] to implement an excitation of the field by voltages instead of currents, the optimal transmission condition can be derived as follows.

Let the field problem be voltage driven, using a GAUSS-SEIDEL scheme. The zero-order Dirichlet conditions are

$$V_F^{k+1} = V_C^{k+1}, \quad (2.3a)$$

$$I_C^{k+1} = I_F^k. \quad (2.3b)$$

The circuit-side transmission condition (2.3b) can be generalised by including both current and voltage in a linear combination, i.e.,

$$I_C^{k+1} - \beta V_C^{k+1} = I_F^k - \beta V_F^k, \quad (2.4)$$

where β is the weighting factor. It is worth noting, that the field’s input condition (2.3a) is not optimised for, because unlike the field problem, an impedance of arbitrary circuits cannot be easily extracted, cf. [4]. In order for the fixed point iteration to converge, the error across iterations has to contract, i.e.,

$$\|I_C^{k+1} - I_C^k\| = |\rho(\beta)| \|I_C^k - I_C^{k-1}\|,$$

where the contraction factor $|\rho(\beta)| \stackrel{!}{<} 1$. Adhering to the same assumptions made in the derivation in [4], an analogous admittance of the MQS problem can be determined, and the contraction factor

$$\rho(\beta) = (\mathbf{I} + \beta \mathbf{x}_P^{-1}(\omega))^{-1} (\beta - \mathbf{Y}(\omega)) \mathbf{x}_P^{-1}(\omega).$$

Optimal convergence is attained for $\rho(\beta) = 0$, such that

$$\beta = \mathbf{Y}(\omega).$$

Thus, the optimized transmission condition of the circuit is given by

$$I_C^{k+1} - \mathbf{Y}(\omega)V_C^{k+1} = I_F^k - \mathbf{Y}(\omega)V_F^k. \quad (2.5)$$

Naturally, using the admittance in our implementation, which is in the time domain, $\mathbf{Y}$ has to be approximated, as pointed out by [4].

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A solution of the resulting system of DAEs can be obtained through known algorithms. Most notably, the GAUSS-SEIDEL (Einzel-schrittverfahren) and GAUSS-JACOBI (Gesamt-schrittverfahren) schemes are common [2]. While the former offers faster convergence, the latter has better parallelizability. The implementation in this work will be carried out using the GAUSS-SEIDEL scheme, i.e.,

$$V_F^{k+1} = V_C^{k+1} \quad \text{and} \quad I_C^{k+1} = I_F^k, \quad (2.6)$$

in the voltage-driven field case.

3 Coupling with Xyce™

To successfully implement dynamic iteration, which is a fixed-point iteration technique, a rollback to repeat the co-simulation step with corrected inputs is required.

3.1 Requirements on the Circuit Solver (Why Xyce?)

3.2 Field Dummy Solver

3.3 Coupling via the Xyce Binary

3.3.1 Restarts and Checkpointing

3.3.2 Interface Condition via Behavioural Source

3.3.3 Coupling Direction

3.4 General External Interface and Its Limitations



4 Benchmark and Verification

Test Suite.



5 Application to General Circuits

Parameterstudien und -Empfehlungen.



6 Conclusions and Outlook

Zusammenfassung der Ergebnisse, Schlussfolgerung und Ausblick. Keine neuen Informationen mehr.



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